

A scenic mountain landscape with green hills, rocky cliffs, and a cloudy sky. The foreground shows a rocky, grassy slope. In the middle ground, there are steep, rocky cliffs and green hills. The background features more distant, hazy mountain ranges under a sky filled with white and grey clouds.

AI & Software Engineering Workshop

Week 2, Day 2: Data and Tokenization

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63 GB of Armenian: from where?

12 sources, one cleaned corpus:

- Wikipedia · Wikisource · Wiktionary · Wikiquote
- CC-100 · HPLT 3.0 · CC-News · CulturaX · mC4 · Glot500
- ARLIS · FineTranslations

Encyclopedias, news, books, the open web: the model will only ever know what's in here.

Garbage in, garbage out

Before training, the corpus was:

- **Cleaned**: boilerplate, menus, broken encoding stripped out
- **Deduplicated**: exact copies *and* near-copies (web text repeats a lot)
- **Prioritized**: high-quality sources first when there's overlap

A model trained on raw internet learns raw internet. Data work is unglamorous and decisive.

Why tokens? Computers eat numbers

- A model can't read "Բարև"; it needs numbers
- **Character-level**: tiny vocabulary, but every word becomes a long sequence
- **BPE (Byte Pair Encoding)**: merge frequent pairs into subword pieces
 - `Հայաստան` → `Հայ` + `աստան` (*illustrative*)
 - Common words = 1 token, rare words = a few pieces
- Our corpus → **~8.3 billion BPE tokens**, 16K-piece vocabulary

Hands-on: think like a tokenizer

1. Train a BPE tokenizer on a small sample file (*instructor-led*)
2. Encode your name. Encode «Երևան». Encode a rare word.
3. Which words survive as one piece? Which shatter?
4. What happens to English words in an Armenian tokenizer?

The vocabulary is a mirror of the training data.

Getting the real data, the smart way

```
python 1_download.py --tokenized-only      # ~10 GB
```

- Pre-tokenized `train_bpe.bin` + `val_bpe.bin` + tokenizer: ready to train
- Re-tokenizing 63 GB yourself: **hours** (even with 16 parallel workers)
- 63 GB × everyone in this room: the venue wifi votes no
- The full from-scratch path is in the repo README: reproduce it at home

Today → Tomorrow

Today: the corpus, the cleaning, and how text becomes numbers.

Tomorrow morning we press the button. 350M parameters start learning Armenian, and keep learning while we sleep.